

```
In[1]:= Clear["Global`*"];
```

0. Readme

1. (x', y', z') are represented by $(x3, y3, z3)$.
2. $V_{xxx}^c, V_{xxy}^c, V_{xxz}^c, V_{yyx}^c, V_{yyy}^c, V_{yyz}^c, V_{zzx}^c, V_{zzy}^c$, and V_{zzz}^c in Table 3 and Eqs. (23-25) are represented by CartesianVxxx, CartesianVxxy, CartesianVxxz, CartesianVyyx, CartesianVyyy, CartesianVyyz, CartesianVzzx, CartesianVzzy, and CartesianVzzz.
3. The integral kernels are considered and the complete expressions should be presented as $G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d u' d \phi' d \lambda'$.

1. Basic settings

```
In[2]:= x = r * Cos[phi] * Cos[lam];
```

```
y = r * Cos[phi] * Sin[lam];
```

```
z = r * Sin[phi];
```

```
In[5]:= Dx = x - x3;
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Dy = y - y3;
```

```
Dz = z - z3;
```

```
In[8]:= dx = -Sin[lam] * Dx + Cos[lam] * Dy;
```

```
dy = -Sin[phi] * Cos[lam] * Dx - Sin[phi] * Sin[lam] * Dy + Cos[phi] * Dz;
```

```
dz = Cos[phi] * Cos[lam] * Dx + Cos[phi] * Sin[lam] * Dy + Sin[phi] * Dz;
```

```
In[11]:= l = FullSimplify[Sqrt[Dx^2 + Dy^2 + Dz^2]];
```

2. Expressions from Table 3 involved in Eqs. (23-25)

```
In[12]:= CartesianVxxx = FullSimplify[-3 * dx * (5 * dx^2 - 3 * l^2) / l^7];
```

```
In[13]:= CartesianVxxy = FullSimplify[-3 * dy * (5 * dx^2 - l^2) / l^7];
```

```
In[14]:= CartesianVxxz = FullSimplify[-3 * dz * (5 * dx^2 - l^2) / l^7];
```

```
In[15]:= CartesianVyyx = FullSimplify[-3 * dx * (5 * dy^2 - l^2) / l^7];
```

```
In[16]:= CartesianVyyy = FullSimplify[-3 * dy * (5 * dy^2 - 3 * l^2) / l^7];
```

```
In[17]:= CartesianVyyz = FullSimplify[-3 * dz * (5 * dy^2 - l^2) / l^7];
```

```
In[18]:= CartesianVzzx = FullSimplify[-3 * dx * (5 * dz^2 - l^2) / l^7];
```

```
In[19]:= CartesianVzzy = FullSimplify[-3 * dy * (5 * dz^2 - l^2) / l^7];
```

```
In[20]:= CartesianVzzz = FullSimplify[-3 * dz * (5 * dz^2 - 3 * l^2) / l^7];
```

3. Test the correctness of Eqs. (23-25)

```
In[21]:= Laplace4 = FullSimplify[CartesianVxxx + CartesianVyyx + CartesianVzzx]
```

```
Out[21]=
```

0

```
In[22]:= Laplace5 = FullSimplify[CartesianVxxy + CartesianVyyy + CartesianVzzy]
```

```
Out[22]=
```

0

```
In[23]:= Laplace6 = FullSimplify[CartesianVxxz + CartesianVyyz + CartesianVzzz]
```

```
Out[23]=
```

0

```
In[24]:= NotebookSave[EvaluationNotebook[]];
```